

LUCA GANDOLFI

Embedded Software & V&V Engineer

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📍 Fidenza, Italy; open to Milan and relocation

PROFESSIONAL SUMMARY

Embedded software engineer with 4+ years in aerospace and defence, specialising in software V&V, test-environment porting, and HW/SW integration for safety-critical systems. Experience with DO-178C/ECSS, VectorCAST, Renode, RISC-V, and SAMRH/PolarFire targets. Strong C/C++, Python, and Linux foundation with hands-on virtual-platform and AI/ML projects.

PROFESSIONAL EXPERIENCE

Software Engineer – Aerospace & Defence

ALTEN Italia

📅 June 2022 – Present 📍 Italy

- Deliver software engineering, verification and validation, and HW/SW integration services for major aerospace and defence clients, including Thales Alenia Space and Leonardo.
- Ported functional, unit, and integration test environments across SAMRH71, SAMRH707, and PolarFire using VectorCAST and Renode.
- Supported repeatable RISC-V firmware verification across simulated and board-oriented workflows.
- Contributed to aerospace software lifecycle documentation under DO-178C and ECSS standards for Thales Alenia Space programmes.
- Refactored a Leonardo OPLS application, improving its structure and maintainability while preserving required behaviour.

ICT Developer – Aerospace

ALTEN Italia

📅 December 2021 – June 2022 📍 Italy

- Performed software testing, verification and validation, and HW/SW integration for aerospace systems.
- Adapted avionics software for LIDAR processing, inter-computer FTP communication, and Linux/Qt applications over AFDX.

University Programming Tutor

Università degli Studi di Parma

📅 2019 and 2021 📍 Parma, Italy

- Taught Go, Python, and C and supported project development for courses in programming fundamentals and programming-language paradigms.

SELECTED TECHNICAL PROJECTS

PolarFire RISC-V Virtual Platform

GitHub

- Built a tested Python PolarFire/RISC-V functional platform with RV64I/M multi-hart execution, ELF loading, MMIO/interrupt models, and GDB remote debugging.

SAMRH Linux Test Environment

GitHub

- Created a Linux host/target test workflow for SAMRH71/707 firmware using C, GCC/Clang, Bash, Make, MPLAB X/XC32, and MDB simulator/hardware paths.

AI/ML and LLM Engineering

GitHub

- Compared BERT/RoBERTa emotion models and explored PySpark parallelisation; built tested Python LLM applications with structured outputs, async APIs, fallback routing, and persistent memory.

EDUCATION

Master's studies in Computer Engineering

Università degli Studi di Parma, Italy 2018 – Present

Bachelor's degree in Computer, Electrical and Telecommunications Engineering

Università degli Studi di Parma, Italy 2013 – 2018

Thesis: *An Overview of Internet of Things: Challenges and Opportunities.*

CORE EXPERTISE

- Software V&V
- Test Automation
- HW/SW Integration
- Functional Testing
- Unit Testing
- Integration Testing
- Safety-Critical Software
- Embedded Systems
- Firmware Validation

STANDARDS & TOOLS

- DO-178C
- ECSS
- VectorCAST
- Renode
- Git
- Linux
- Bash
- MPLAB X
- XC32
- GCC/Clang
- Make

PLATFORMS

- RISC-V
- PolarFire SoC
- SAMRH71
- SAMRH707
- STM32
- FreeRTOS

PROGRAMMING

Primary

C, C++, Python, Bash

Additional

Java, Go, SQL, JavaScript, Assembly

PROTOCOLS & FRAMEWORKS

- CAN
- I2C
- AFDX
- OPC UA
- FTP
- Qt
- PySpark
- TensorFlow
- BERT
- FastAPI

LANGUAGES

Italian – Native
English – Professional proficiency

PORTFOLIO

Portfolio repository
Technical projects repository